

AIGR Focused Review — *Drosophila melanogaster* Hsp22 (P02515): conditional nuclear localization

Focus type: function_assignment · **Hypothesis slug:** conditional-nuclear-localization **Target GO term under scrutiny:** GO:0005634 *nucleus* (CC), currently annotated **IBA** for P02515.

Executive Judgment

Verdict: OVER-ANNOTATED (nucleus IBA) / core location = mitochondrial matrix (strongly supported).

The proposition that Hsp22 has a *demonstrated* nuclear localization is **not supported by any Hsp22-specific experimental evidence**. The `nucleus` (GO:0005634) annotation on P02515 rests **solely on a phylogenetic IBA** (`is_active_in`, GO_REF:0000033, propagated from PANTHER sHSP tree PTHR45640). The only experimentally supported compartment is the **mitochondrial matrix** (GO:0005759), which carries a direct **IDA** annotation from primary work

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P 10896659

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and is corroborated by ≥ 4 additional independent sources. Sequence analysis is consistent: P02515 has a positively charged, amphipathic N-terminal segment typical of a mitochondrial targeting sequence and **no canonical NLS motif**.

The seed hypothesis correctly warns against conflating (i) family-level IBD inference, (ii) ambiguous historical nuclear-fraction bands, (iii) Hsp23 paralog immunofluorescence, and (iv) same-nickname mammalian HSPB8 / plant Hsp22. When those confounders are removed, **no residual direct evidence for Hsp22 nuclear import remains**.

Calibrated conclusion: *Absence of evidence, not demonstrated presence.* The nucleus IBA should be treated as paralog/family over-annotation. This is a **lead for curator action** (remove or do-not-accept the nucleus CC), while retaining mitochondrial matrix as the core CC. Caveat: IBA is a legitimate GO_Central pipeline call; removal requires either a curator NOT qualifier / annotation-review, or an upstream PANTHER IBD reassessment.

Evidence Matrix

Citation	Evidence type	Stance	Claim tested	Key finding
P 10896659 (Morrow et al. 2000)	direct assay / localization	supports (mito)	Endogenous Hsp22 compartment	S2 mitochondrial colocalization after 35 °C/1 h + 2 h recovery; matrix fractionation + protease protection in transfected cells; N- terminal import mapping
P 19948727 (Wadhwa et al. 2010)	localization / review	supports (mito)	Hsp22 compartment	"DmHsp22 ... is localized in the mitochondrial matrix"
P 19420297 (Yang & Tower 2009)	localization / review	supports (mito)	Hsp22 compartment	"Hsp22 ... localizes to the mitochondrial matrix"
P 15491684 (Bhole et al. 2004)	localization / review	supports (mito)	Hsp22 compartment	"localizes to the mitochondrial matrix"
P 26155908 (Morrow et al. 2016, review)	review	supports (mito)	Intramitochondrial sHSP	"one of the members of the family to be localized inside mitochondria"
GO_REF:0000033 — PANTHER PTHR45640 IBA	computational (IBA)		Ancestral sHSP activity location	nucleus & cytoplasm

Citation	Evidence type	Stance	Claim tested	Key finding
		competing (nucleus/ cyto)		inferred is_active_in via phylogeny (leaf PTN000163333 under nucleus IBD PTN000897708)
DOI:10.1016/0012-1606(80)90320-6 (Dev Biol 1980)	localization (historical)	qualifies (nucleus)	HSPs in nuclear/ nucleolar fractions	22/23/26/27 kDa bands in nuclear/ chromatin/ nucleolar preparations after 37 °C
DOI:10.1139/g86-152 + UNIGE(6801431) — Hsp23 IF	localization	competing (paralog)	Nuclear/nucleolar sHSP IF	Hsp23 nuclear/ nucleolar signal; authors state Hsp26/ Hsp27 cross-reactivity cannot be excluded
Sequence P02515 (this run)	computational / structural	supports (mito)	Targeting signals	174 aa; sHSP (ACD) domain 44–154; N-term net-positive, amphipathic MTS-like; no monopartite/

Citation	Evidence type	Stance	Claim tested	Key finding
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bipartite NLS
motif

Provenance: GO annotations retrieved live from QuickGO (`geneProductId=P02515`, aspect `cellular_component`, 3 hits). Sequence pulled from UniProt REST (`P02515.json`). Both executed in-run; console output is in the iteration log. CSV export to disk was blocked by the sandbox (read-only), so tables are embedded here.

Computed GO decision table (from QuickGO, iteration 2)

GO_ID	term	evidence	reference	assigned_by	qualifier	curation lead
GO:0005759	mitochondrial matrix	IDA (ECO:0000314)	P 10896659	FlyBase	located_in	RETAIN (core CC)
GO:0005634	nucleus	IBA (ECO:0000318)	GO_REF:0000033	GO_Central	is_active_in	REMOVE / do-not-accept (lead)
GO:0005737	cytoplasm	IBA (ECO:0000318)	GO_REF:0000033	GO_Central	is_active_in	REVIEW (weak; likely non-core)

GO Curation Implications

- **GO:0005759 mitochondrial matrix (CC) — RETAIN as core.** Direct IDA, multiply corroborated. This is the primary, gene-specific location.
- **GO:0005634 nucleus (CC) — candidate REMOVE / do-not-accept (lead).** Purely IBA; conflicts with a strong experimental matrix localization plus an MTS-bearing, NLS-less sequence. There is no Hsp22-specific assay of nuclear import. For a mitochondrially-targeted, derived family member, the family IBD is a poor guide.

- **GO:0005737 cytoplasm (CC) — REVIEW (weak).** Also IBA-only. A transient/cytosolic pre-import pool is biologically plausible for any nucleus-encoded mitochondrial protein, but it is not experimentally documented for Hsp22 specifically; treat as non-core at best.
- **MF/BP note (orientation, not the focus):** the well-supported activities are *unfolded protein binding / holdase chaperone* (MF) and roles in the mitochondrial unfolded protein response and proteostasis (BP). Per the seed instruction, these established holdase/longevity annotations are not re-litigated here.
- Recommendation avoids "protein binding": the informative supported CC term is **mitochondrial matrix**.

Mechanistic Scope

The immediate, direct molecular activity of Hsp22 is ATP-independent **holdase chaperone** action (α -crystallin domain, residues 44–154) that binds unfolding substrates and retards their aggregation. Its direct cellular context is the **mitochondrial matrix**, where it participates in mitochondrial proteostasis and the UPR^{mt}. "Nuclear localization" would be a distinct *cell-biological* claim about protein trafficking — not a downstream phenotype but a direct localization statement — and it is precisely that direct claim that lacks Hsp22-specific evidence. Longevity, oxidative-stress resistance, and UPR^{mt} effects are **downstream physiological consequences** and are not evidence for a nuclear pool.

Conflicts and Alternatives

1. **Paralog confusion (primary alternative).** Nuclear/nucleolar sHSP signals in *Drosophila* are documented for **Hsp23, Hsp26, Hsp27** — not Hsp22. The 1986 Hsp23 study explicitly could not exclude Hsp26/Hsp27 antibody cross-reaction. IBD propagation across PTHR45640 lets these paralogs' inferred nuclear activity flow onto Hsp22.
2. **Historical band ambiguity.** The 1980 nuclear-fraction data pre-date individual sHSP cloning; the 22/23/26/27 kDa cluster co-migrates and was not resolved with an Hsp22-specific reagent, so a "22 kDa nuclear band" is not equivalent to "Hsp22 in the nucleus."

3. **Same-nickname trap.** Mammalian **HSPB8** (human "Hsp22", cytoplasmic/nuclear in some contexts) and **plant Hsp22** (ER/organellar) are non-orthologous to fly Hsp22; their localizations must not transfer.

4. **Fractionation $\neq$ nucleus.**

P 3109982

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whole-insect soluble/particulate fractionation does not resolve a nuclear compartment.

5. **Sequence signal points the other way.** MTS-like N-terminus + no NLS argues against constitutive nuclear import.

No evidence found that positively supports a real, reproducible Hsp22 nuclear pool.

Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
Is there any modern Hsp22-specific nuclear imaging?	PubMed searches for Hsp22 localization; QuickGO annotations	Would be the only way to upgrade nucleus beyond IBA	GFP-Hsp22 or validated-antibody confocal with mito/nuclear co-stains under stress + recovery time course
Exact PTHR45640 membership / which leaves carry the nucleus IBD	PANTHER REST endpoint (HTTP 404, not accessible in-run)	Confirms the IBD is driven by non-mitochondrial paralogs	Inspect PANTHER tree PTHR45640 node PTN000897708 experimental donors in the GO_Central pipeline
Full text of the 1980 Dev Biol paper	Abstract/seed only (not indexed under tried queries)	Band identity, fraction purity, EM autoradiography specificity, later reassessments	Retrieve DOI:10.1016/0012-1606(80)90320-6 full text and any follow-up reassessment
Transient pre-import cytosolic pool	Sequence + IBA only	Distinguishes GO:0005737 relevance	Pulse-chase import assay / isolated-mitochondria import with mutants

Discriminating Tests

1. **Endogenous GFP-tag or validated antibody confocal** in S2 cells and adult tissues, with MitoTracker + DAPI, across a heat-shock/recovery time course — the single most decisive test for a real vs. artifactual nuclear pool.
2. **Digitonin fractionation / super-resolution** to separate matrix from any nuclear signal; protease protection already supports matrix.

3. **NLS/MTS swap constructs:** delete N-terminal MTS and test for nuclear redistribution to probe latent import capacity (addresses the seed's "import-mutant mapping does not prove universal nuclear exclusion" caveat).
4. **Isoform/paralog-clean reagents:** ensure antibodies do not cross-react with Hsp23/26/27 (the historical failure mode).
5. **PANTHER tree audit:** confirm whether the nucleus IBD donor annotations come exclusively from cytosolic/ nuclear sHSP paralogs, justifying a taxon/branch-specific NOT for the mitochondrial clade.

Curation Leads (require curator verification)

- **Action lead:** For P02515, **do not accept / consider removing GO:0005634 (nucleus)** as an over-propagated IBA; **review GO:0005737 (cytoplasm)** as weak/non-core; **retain GO:0005759 (mitochondrial matrix)** as the core experimental CC.

- **Candidate anchor reference (retain/strengthen mito):**

P 10896659

— verify snippet on S2 mitochondrial colocalization + matrix fractionation/protease protection.

- **Corroborating references (mito):**

P 19948727

("localized in the mitochondrial matrix");

P 19420297

("localizes to the mitochondrial matrix");

P 15491684

("localizes to the mitochondrial matrix").

- **Competing/into-notes references (paralog nuclear signal, do NOT attribute to Hsp22):** DOI:10.1139/g86-152 and the Hsp23 salivary-gland IF; DOI:10.1016/0012-1606(80)90320-6 historical nuclear-fraction bands.
- **Suggested curator question:** "Is any donor annotation under PANTHER PTHR45640 nucleus IBD (PTN000897708) based on an experimentally nuclear *mitochondrial* sHSP, or

exclusively cytosolic/nuclear paralogs?" If the latter, a branch-specific NOT for the mitochondrial clade is justified.

- **Suggested experiment:** endogenous-level GFP-Hsp22 confocal time course with mito+nuclear co-stains.
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Limitations

- No wet-lab data generated; assessment is literature + public-annotation + sequence-heuristic based.
 - PANTHER family membership could not be pulled programmatically (endpoint 404); the IBD provenance statement is taken from the seed and the QuickGO evidence codes, not an independent tree parse.
 - The 1980 Dev Biol full text was not retrieved in-run; its treatment relies on the publisher abstract / seed description. NLS detection used regex motif heuristics, not a trained predictor.
 - "Absence of evidence" for nuclear Hsp22 is not proof of absence; a rare/conditional nuclear pool cannot be formally excluded, which is why the recommendation is framed as a curator lead.
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